



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

B.Sc. DEGREE EXAMINATION

COLLEGE: Science & Technology

SESSION: 2024/2025

COURSE CODE: COS 221

COURSE TITLE: Object-Oriented Programming (Java)

INSTRUCTIONS: Attempt any four (4) questions

DEPARTMENT: Computer & Info. Science

SEMESTER: Omega

CREDIT UNIT: 3

TIME: 3 HOURS

Question One

17.5 Marks

A. Define Method signature, Method overloading and Void Method.

[3 marks]

B. Debug this code snippet, list and correct the errors.

[3 marks]

```
public class SquareCalc {  
    public static void main(String[] ) {  
        int num = 5;  
        double result = square(num)  
        System.out.println("Square is: " + result);  
    }  
  
    public static int square(int) {  
        return n * n;  
    }  
}
```

C. Dry run this code snippet with the following values: 3,4,5,7.

[4 marks]

```
public static int sumSeries(int i) {  
    int sum = 0;  
    for (int j = 1; j <= i; j++) {  
        sum += j / (j + 1);  
    }  
    return sum;  
}
```

D. Write a program that *overloads* a method named area to compute:

[7.5 marks]

- The area of a circle
- The area of a rectangle
- The area of a triangle.

Call these methods and *print out* their values in the main method.

Question Two

17.5 Marks

A. Define the following: Dynamic array initialisation, the length property of arrays, and the out-of-bounds exception in arrays

[3 marks]

B. Debug this code snippet, list and correct the errors.

[3 marks]

```
public class CopyArray {
    public static void main(String[] args) {
        int[] a = {2, 4, 6, 8};
        int b;
        System.arraycopy(a, 0, b, 0, a.length);
        for(int i = 0; i <= b.length; i) {
            System.out.println(b[i]);
        }
    }
}
```

C. Dry run this code. State the output for each iteration and state the final output

[4 marks]

```
public class Array {
    public static void main(String[] args) {
        int[] nums = {2, 4, 6, 8};
        int x = nums[0];
        for (int i = 1; i < nums.length; i++) {
            x = x - nums[i];
        }
        System.out.println(x);
    }
}
```

D. Write a program that does the following:

[7.5 marks]

- i. *Prompts* the user to input 10 values into an array
- ii. Implement two methods
 - a. A method to *find*, *count* and *display* the number of even
 - b. Another method to *sum* and *display* the odd numbers.
- iii. In the main method, use a for loop to *find* and *display* the maximum number.

Question Three

17.5 Marks

A. Define the following: PrintWriter, Files.write() method and try-with-resources block.

[3 marks]

B. Debug this code snippet, list and correct the errors.

[3 marks]

```
java.io.FileWriter;

public class FileTest {
    public static void main(String[] args) {
        FileWriter writer = FileWriter("note.txt");
        writer.write("Hello from Java!");
        writer.close
    }
}
```

C. What is the output of the data.txt file?

[4 marks]

```
import java.io.*;

public class PrintWriterExample {
    public static void main(String[] args) throws IOException {
        PrintWriter pw = new PrintWriter("data.txt");
        pw.print("Age: ");
        pw.println(25);
        pw.printf("Score: %.2f", 89.456);
        pw.close();
    }
}
```

D. Write a program that does the following:

[7.5 marks]

- i. Create a file named `students.txt` if it does not exist.
- ii. Uses `PrintWriter` to write the following student records to the file in a formatted table

Name	Age	GPA
John	20	3.75
Lilian	21	3.95
Ken	19	3.65

- iii. Use `printf()` for proper column alignment
- iv. Ensure the file is appended to if it already exists.
- v. Handle possible exceptions properly

Question Four

17.5 Marks

A. Describe the following statements: Decision-making, Looping and Branching.

[3 marks]

B. Debug this code snippet, list and correct the errors.

[3 marks]

```
int count = 1
float output = "";

while(count <=5); {
    output = output + count + " ";
    Count
}

System.out.println(output);
```

C. State the output of each iteration and the final sum.

[4 marks]

```

int sum = 0;
int[] inputs = {3, 4, -1, 2};

for(int i = 0; i < inputs.length; i++) {
    if(inputs[i] < 0)
        break;
    sum += inputs[i];
}
System.out.println("Sum = " + sum);

```

D. Write a program to define a class called `TemperatureCheck` with the following specifications: [7.5 marks]

- i. Data field (state): double temperature.
- ii. A constructor method to *set* the temperature
- iii. A method called `evaluateTemperature()` that *uses* if-else statements to output:
 - a. "Too Cold" if temperature < 15
 - b. "Normal" if the temperature is between 15 and 30 (inclusive)
 - c. "Too Hot" if temperature > 30.
- iv. In your main method, *prompt* the user to enter a temperature value, *create* an object of the class, and *display* the evaluation result. *Use* `JOptionPane` to accept inputs.

Question Five

17.5 Marks

- A. Describe the following statements: Inheritance, Polymorphism, and Method Overriding. [3 marks]
- B. *Write* a method called `countNegatives` that accepts a single-dimensional array of integers and returns the number of negative values in the array. You do not need to write the full class or main method, just the method definition. [3 marks]
- C. Identify the errors in this code snippet and state the final output after correction. [4 marks]

```

public class Main {
    public static void main(String[] args) {
        int nums = {10, 20, 30};
        System.out.println("Middle value is: " + nums[3]);
        int sum = 0;
        for(int i = 0; i <= nums.length; i) {
            sum += nums[i];
        }
        System.out.println("Sum is: " + );
    }
}

```

D. Write a program using inheritance and method overriding as follows: [7.5 marks]

- i. *Create* a superclass called `Vehicle` with the following:
 - a. An instance variable `speed` (int)
 - b. A constructor that initialises `speed`
 - c. A method `displayInfo()` that *prints* "Vehicle speed: <speed> km/h"

- ii. Create a subclass called Bicycle that *overrides* the `displayInfo()` method to print: "Bicycle speed: <speed> km/h. This is an eco-friendly ride."
- iii. In the main method:
 - a. Create an object of class Vehicle with a speed of 80
 - b. Create an object of class Bicycle with a speed of 25
 - c. Call `displayInfo()` on both objects to demonstrate polymorphism.

Question Six

17.5 Marks

A. Write a code snippet to demonstrate the following loop types:

[3 marks]

- i. A for loop that prints numbers 1 to 5
- ii. A while loop that prints numbers from 5 to 1
- iii. A do-while loop that prints only the number 10 once.

B. What will be the return value of `compute(13)`? Show the steps to explain your answer.

[3 marks]

```
public static int compute(int x) {
    int count = 0;
    while (x > 0) {
        if (x % 2 == 1) count++;
        x = x / 2;
    }
    return count;
}
```

C. Debug this code snippet. List the errors and state the correct output.

[4 marks]

```
public class EvenFinder {
    public static void main(String args) {
        printEven(10.0)
    }

    public static void printEven(int) {
        for (int i = 0; i <= n; i++);
        if (i % 2 == 0)
            System.out.print(i + " ");
    }
}
```

D. Write a program that includes a method called `hasEqualAdjacentPairs(int n)` which:

[7.5 marks]

- i. Returns true if the number `n` (a positive integer) contains at least two pairs of equal adjacent digits.
- ii. Returns false otherwise.
- iii. You must:
 - a. Write the method `hasEqualAdjacentPairs(int n)`
 - b. Write the `main()` method to *test* at least two cases

Example

- 112233 - true (pairs: 11, 22, 33). More than two adjacent pairs.
- 1223 - true (pairs: 22). Only one pair - return false

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<i>[Signature]</i>
HEAD OF DEPARTMENT
Date: 14/11/25